AN ORDER-49 PARTITION REFUTES THE HEAVY-CHOPPED-RECTANGLE INTERVAL CONJECTURE OF GOTTLIEB, KRNC AND MURŠIČ

ABSTRACT. In 1-PNim a move deletes a positive number of rows, or a positive number of columns, from the Young diagram of a single partition; a position is *heavy* when its Sprague–Grundy value equals the length of the longest play from it. Gottlieb, Krnc and Muršič conjectured that whenever the rectangle $[(a+1)^{b+1}]$ is heavy, every λ with $[a+1, a, \dots, a-b+1] \le \lambda \le [(a+1)^{b+1}]$ in Young's lattice is heavy. We refute this. For $(a,b) = (10,4)$ the rectangle $[11^5]$ is heavy, yet $\lambda = [11, 11, 11, 8, 8]$, which lies strictly between the two endpoints, has $\mathrm{sg}(\lambda) = 10$ where the conjecture demands $a + b + 1 = 15$. The certificate is a mex table over the 45 options of λ, printed here in full. No counterexample has order less than 49, and exactly one other has order 49.

1. The game and the conjecture

Let $\lambda = [\lambda_1, \dots, \lambda_r]$ be a partition, written as a weakly decreasing tuple of positive parts, and let $\lambda' = [\lambda'_1, \dots, \lambda'_{\lambda_1}]$ be its conjugate. In the game 1-PNim of Gottlieb, Krnc and Muršič [1], the moves from λ are

(1) $\lambda \to [\lambda_{i_1}, \dots, \lambda_{i_\ell}]$ for a proper subsequence $i_1, \dots, i_\ell$ of $1, \dots, r$; and
(2) $\lambda \to [\lambda'_{i_1}, \dots, \lambda'_{i_\ell}]'$ for a proper subsequence $i_1, \dots, i_\ell$ of $1, \dots, \lambda_1$,

that is, a move deletes a positive number of rows, or a positive number of columns, and the surviving rows (columns) merge. The empty partition is the unique terminal position. We write sg for the Sprague–Grundy value under normal play, so that $\mathrm{sg}(\lambda) = \mathrm{mex}\{\mathrm{sg}(\mu) : \lambda \to \mu\}$ and $\mathrm{sg}([\,]) = 0$.

The longest play from a nonempty λ has length $L(\lambda) := \lambda_1 + r - 1$, whence $\mathrm{sg}(\lambda) \le L(\lambda)$ [1, Prop. 2.6]; the position λ is called *heavy* when equality holds. Two families of heavy positions are relevant here. First,

$$\mathrm{sg}\big([c^r]\big) = \big((r-1) \oplus (c-1)\big) + 1 \tag{1}$$

for all positive r, c [1, Thm. 1.1], where $\oplus$ is the Nim-sum, so $[c^r]$ is heavy exactly when $(r-1) \oplus (c-1) = (r-1) + (c-1)$, i.e. when $\binom{c+r-2}{r-1}$ is odd. Second, the staircase $\mathrm{st}_{c,r} := [c, c-1, \dots, c-r+1]$ is heavy for all $c \ge r \ge 1$ [1, Prop. 4.4]. Young's lattice is the order in which $[\lambda_1, \dots, \lambda_\ell] \le [\mu_1, \dots, \mu_m]$ means $\ell \le m$ and $\lambda_j \le \mu_j$ for $j = 1, \dots, \ell$.

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The conjecture we refute observes that the two endpoints of a certain interval of Young's lattice are heavy for the two reasons just quoted, and asks whether everything between them is heavy as well. It carries no printed number: it appears in the open-problems section of [1] under the source's own label `conj:heavychoppedrect`, which is the only stable handle the source provides, and we therefore quote it verbatim from the source file of the e-print of `2506.04991v1`:

```
\begin{conj}\label{conj:heavychoppedrect}
    Let $a,b\in\ZZ$ be such that $\br{(a+1)^{b+1}}$ is heavy.
      %a\ge b$ and $a\oplus b=a+b$.
    Then any $\lambda$ satisfying
    $\br{a+1,a,\dots, a-b+1}\le\lambda\le \br{(a+1)^{b+1}}$ is heavy.
    %that is, $\sgpnim(\lambda)=a+b+1$.
\end{conj}
```

In display, therefore:

Conjecture 1 (Gottlieb–Krnc–Muršič)**.** Let $a, b \in \mathbb{Z}$ be such that $[(a+1)^{b+1}]$ is heavy. Then any λ satisfying $[a+1, a, \dots, a-b+1] \le \lambda \le [(a+1)^{b+1}]$ is heavy.

Three remarks fix the reading, and none of them creates an escape. *First*, two clauses of the hypothesis are commented out in the source and so do not appear in the compiled paper: $a \ge b$ and $a \oplus b = a + b$. The second is not a restriction at all: by (1) the printed hypothesis "$[(a+1)^{b+1}]$ is heavy" says $\binom{a+b}{b}$ is odd, which by the parity criterion for binomial coefficients [2] is exactly $a \wedge b = 0$, i.e. $a \oplus b = a + b$. The first is a real restriction, and it is needed for the lower endpoint to be a partition at all: if $b > a$ then $[a+1, a, \dots, a-b+1]$ has nonpositive entries. The counterexample below has $a = 10 \ge 4 = b$, so it refutes Conjecture 1 under either reading. *Second*, both endpoints have first part $a+1$ and exactly $b+1$ rows, so by Young's lattice every λ in the interval has $\lambda_1 = a+1$ and $r = b+1$ as well; hence $L(\lambda) = a+b+1$ for every member, and "λ is not heavy" and "$\mathrm{sg}(\lambda) < a+b+1$" are the same statement, which is what the third commented line records. *Third*, the two endpoints really are heavy under the hypothesis: the upper one by assumption, the lower one because it is $\mathrm{st}_{a+1,b+1}$.

2. The counterexample

Theorem 2. *Let* $(a, b) = (10, 4)$ *and* $\lambda = [11, 11, 11, 8, 8]$*, a partition of* 49*. Then*

(1) $[11^5]$ *is heavy, so the hypothesis of Conjecture 1 holds at* $(a, b) = (10, 4)$*;*
(2) $[11, 10, 9, 8, 7] \le \lambda \le [11^5]$ *in Young's lattice, and* λ *is neither endpoint; and*
(3) $\mathrm{sg}(\lambda) = 10$*, whereas* $L(\lambda) = a + b + 1 = 15$.

Consequently λ is not heavy and Conjecture 1 is false.

Proof. (1) $\binom{a+b}{b} = \binom{14}{4} = 1001$ is odd, equivalently $10 \wedge 4 = 0$; and (1) gives $\mathrm{sg}([11^5]) = ((5-1) \oplus (11-1)) + 1 = (4 \oplus 10) + 1 = 14 + 1 = 15 = 11 + 5 - 1 = L([11^5])$.

(2) The lower endpoint is $\mathrm{st}_{11,5} = [11, 10, 9, 8, 7]$, and $11 \le 11$, $10 \le 11$, $9 \le 11$, $8 \le 8$, $7 \le 8$ on five rows; the upper endpoint dominates λ because every part of λ is at most 11 and both have five rows. Neither endpoint equals λ.

(3) We first list the options. The rows of λ are $11, 11, 11, 8, 8$, so a move of type (1) keeps i of the three 11s and j of the two 8s with $0 \le i \le 3$, $0 \le j \le 2$ and $(i, j) \ne (3, 2)$: that is $12 - 1 = 11$ options $[11^i, 8^j]$. The conjugate is $\lambda' = [5^8, 3^3]$, so a move of type (2) keeps p of the eight 5s and q of the three 3s with $0 \le p \le 8$, $0 \le q \le 3$ and $(p, q) \ne (8, 3)$, giving $[5^p, 3^q]' = [(p + q)^3, p^2]$ after deleting zero parts: that is $36 - 1 = 35$ options, pairwise distinct because (p, q) is recovered from $(p + q, p)$. A partition occurring in both lists with a positive part would need either five parts in the pattern $3 + 2$, which among the row options forces the excluded $(i, j) = (3, 2)$, or three equal parts $[q^3]$ with $q \le 3$ matching $[11^3]$ or $[8^3]$, which is impossible. So the two lists meet only in $[\,]$ and

$$\#\{\mu : \lambda \to \mu\} = 11 + 35 - 1 = 45.$$

Next, no option can carry the value 15. Every option loses at least one row or at least one column and gains neither, so $\mu_1 + \ell(\mu) \le \lambda_1 + r - 1 = 15$ for every option μ, i.e. $L(\mu) \le 14$; by $\mathrm{sg} \le L$ this gives $\mathrm{sg}(\mu) \le 14$. (This step is the source's own, inside the proof of [1, Prop. 2.6].) Hence

$$\begin{aligned} \lambda \text{ is heavy } &\iff \mathrm{sg}(\lambda) = 15 \\ &\iff \text{the options of } \lambda \text{ realise every value } 0, 1, \ldots, 14, \end{aligned}$$

and exhibiting one value in $\{0, \ldots, 14\}$ realised by no option suffices. Table 1 lists all 45 options with their Sprague–Grundy values. The value set is

$$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} \cup \{11, 12, 13, 14\},$$

so 10 is realised by no option while $0, \ldots, 9$ all are, and $\mathrm{sg}(\lambda) = \mathrm{mex} = 10$. Since $10 < 15$, λ is not heavy. ☐

3. Least order

Call a pair (a, b) with $a \ge b \ge 0$ and $a \wedge b = 0$ *admissible*, so that Conjecture 1 applies to it.

Proposition 3. *No member of the interval of an admissible cell of order less than* 49 *fails to be heavy, and exactly two of order* 49 *do:* $[11, 11, 11, 8, 8]$ *in the cell* $(10, 4)$*, with* $\mathrm{sg} = 10$ *against a required* 15*, and* $[9, 9, 7, 7, 6, 5, 4, 2]$ *in the cell* $(8, 7)$*, with* $\mathrm{sg} = 3$ *against a required* 16*.*

μ		sg	μ		sg	μ		sg
$[10, 10, 10, 8, 8]$		14	$[7, 7, 7, 5, 5]$		11	$[4, 4, 4, 3, 3]$		8
$[10, 10, 10, 7, 7]$		14	$[6, 6, 6, 6, 6]$	$*$	2	$[4, 4, 4, 2, 2]$		8
$[9, 9, 9, 8, 8]$		13	$[11, 11, 8]$		13	$[8, 8]$	$*$	7
$[9, 9, 9, 7, 7]$		13	$[7, 7, 7, 4, 4]$		11	$[3, 3, 3, 3, 3]$	$*$	7
$[11, 11, 11, 8]$		14	$[6, 6, 6, 5, 5]$		2	$[4, 4, 4, 1, 1]$		8
$[8, 8, 8, 8, 8]$	$*$	4	$[11, 8, 8]$		13	$[3, 3, 3, 2, 2]$		7
$[9, 9, 9, 6, 6]$		13	$[6, 6, 6, 4, 4]$		2	$[3, 3, 3, 1, 1]$		7
$[8, 8, 8, 7, 7]$		12	$[5, 5, 5, 5, 5]$	$*$	1	$[11]$	$*$	11
$[11, 11, 8, 8]$		14	$[6, 6, 6, 3, 3]$		2	$[2, 2, 2, 2, 2]$	$*$	6
$[8, 8, 8, 6, 6]$		12	$[5, 5, 5, 4, 4]$		1	$[3, 3, 3]$	$*$	1
$[7, 7, 7, 7, 7]$	$*$	3	$[11, 11]$	$*$	12	$[2, 2, 2, 1, 1]$		6
$[8, 8, 8, 5, 5]$		12	$[5, 5, 5, 3, 3]$		1	$[8]$	$*$	8
$[7, 7, 7, 6, 6]$		4	$[4, 4, 4, 4, 4]$	$*$	8	$[2, 2, 2]$	$*$	4
$[11, 11, 11]$	$*$	9	$[5, 5, 5, 2, 2]$		9	$[1, 1, 1, 1, 1]$	$*$	5
$[11, 8]$	$\dagger$	12	$[1, 1, 1]$	$*$	3	$[]$	$*$	0

Table 1. All 45 options of $\lambda = [11, 11, 11, 8, 8]$, listed in the order of decreasing size within each column. The value 10 does not occur, and $0, \ldots, 9$ all do, so $\mathrm{sg}(\lambda) = 10$. A star marks the 17 entries whose value is forced without computation — the 16 rectangles, by (1), together with $[]$; a dagger marks the one further entry forced by the two-row formula of [1, Prop. 4.9]. The 17 starred values already comprise $\{0, \ldots, 9, 11, 12\}$, so $\mathrm{sg}(\lambda) \geq 10$ needs nothing beyond a Nim-sum; the decisive half — that no option carries 10 — rests on the remaining 27 values.

The search range is finite for the following reason. The least-order member of the cell (a, b) is its lower endpoint $\mathrm{st}_{a+1,b+1}$, of order $(b+1)(2a+2-b)/2$, so a cell can hold a member of order at most 49 only if that quantity is at most 49; this leaves exactly 78 admissible cells, namely all 49 with $b = 0$, then $b = 1$ with a even and $a \leq 24$, $b = 2$ with $a \in \{4, 5, 8, 9, 12, 13, 16\}$, $b = 3$ with $a \in \{4, 8, 12\}$, $b = 4$ with $a \in \{8, 9, 10\}$, $b = 5$ with $a = 8$, $b = 6$ with $a \in \{8, 9\}$, and $b = 7$ with $a = 8$ — every admissible cell with $b \geq 8$ has least order at least 117. The 49 cells with $b = 0$ need no work, their interval being the single point $[a + 1]$, heavy by hypothesis. Proposition 3 is then the result of enumerating, in each of the remaining 29 cells, every member of order at most 49 and evaluating sg; unlike Theorem 2 it rests on that computation and not on a hand-checkable certificate.

Nothing here refutes any of the heaviness results proved in [1], and no cell is claimed free of counterexamples above order 49.

4. Reproduction

Everything in Theorem 2 except the 27 unstarred entries of Table 1 can be checked with a pen: $\binom{14}{4} = 1001$, the Nim-sum $4 \oplus 10 = 14$, five comparisons of parts for the membership, the combinatorial count $11 + 35 - 1 = 45$, and the closed form (1) on the 16 rectangular options. Those checks alone give $\mathrm{sg}(\lambda) \geq 10$, since the starred values already realise $0, \ldots, 9$. To close the argument by hand one must additionally confirm that none of the 27 remaining options has value 10; each of those is a mex over its own option set, and the whole recursion below λ visits 194 distinct partitions.

The program `verify.py` accompanying this note re-derives every computational claim above from the move rule alone, in exact integer arithmetic, reading the objects as printed here; run it as `python3 verify.py` with Python 3.9 or later and the standard library only. It prints the full 45-row table of Table 1, the sweep behind Proposition 3, and one line per check, and exits nonzero unless every check passes. The recorded run `verify.output.txt` reports 87 passing checks and closes with a statement of what it does not cover, in particular that the closed forms quoted above and the source's two appendix tables are transcribed from the e-print rather than checked against it. Neither file is needed in order to redo the calculation of Theorem 2, since every object it consumes is printed above.

References

[1] E. Gottlieb, M. Krnc, and P. Muršič, *Nim on integer partitions and hyperrectangles*, preprint, 5 June 2025. arXiv:2506.04991v1. Numbered statements are cited as they appear in the compiled `v1` preprint; each is also stated in full above.

[2] N. J. Fine, *Binomial coefficients modulo a prime*, Amer. Math. Monthly **54** (1947), no. 10, 589–592. jstor:2304500.